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Professional Ethics

Activity 3: Analyze an Ethical Issue

For this activity I have selected Option 3: Should We Get Rid of Mosquitos?

Summary:

Mosquitoes cause many deaths per year, and many more illnesses, this can be solved by controlling the population size of mosquitoes. One method of doing so would be to alter the genes of a control group in a lab to ensure that, over time, the targeted species of mosquito would eventually go sterile. This has been shown to be very effective at spreading far and could help control the population of disease spreading mosquitoes. However, the overall ecological impact of this is still largely unknown with a few alternative solutions, such as: ensuring we are able to reverse the gene if needed due to unforeseen effects, or even enhancing the mosquitoes to be resistant to the parasites that are the origin of the diseases being spread instead. But this has large impacts on the human life in the affected countries, as well as unknown impacts to the ecological environment. A wrong move can either continue to see increasing numbers of humans die to the diseases spread, or eradicate portions of our delicate ecosystems.

Analysis:

1. Boundaries of the issue:
    a. Mosquitoes cause 597,000 deaths per year and many more illnesses (CDC, 2026). This extends to a huge number of humans across the globe in 83 different countries.
    b. There is a large possibility of damage. Entire ecosystems could eventually be put out of order and could have impacts on many species who rely on the population to survive. This has adverse reactions that is difficult to predict and can easily impact human life as the ecosystem attempts to adapt to the changing food chain.
    c. This would be both an ethical and a legal decision as this could affect local wildlife that has laws on preservation, as well as impacts to the food chain that is difficult to determine the scope of.
2. Research the facts and identify stakeholders:
    a. Stakeholder #1: Humans.
    Mosquitoes cause 597,000 deaths per year and many more illnesses (CDC,

2026). This extends to a huge number of humans across the globe in 83 different countries.

   b. Stakeholder #2: Ecology.
      Mosquitos are a key food source for many animals, including bats, dragonflies, Frogs/Toads, Tadpoles (consumes their larvae), some species of fish, and some birds. Some mosquitoes even pollinate our plants. (Bhattacharya et al., 2016)
   c. Further research is needed on individual species of mosquitoes and their overall impact to our planet's ecological health.
   d. Consultancy with ecologists and other stakeholder representatives (people from the highest affected communities, both humans who suffer from diseases and humans from ecologies that rely heavily on mosquito populations would be required to make informed decisions.

3. Identify possible solutions and evaluate:
   a. First possible solution is to target specific species of mosquitoes only and have only that specific species carry a killing gene that would eventually make them sterile, eradicating only the problematic species. This would be the best utilitarian perspective with the most human lives saved the fastest.
   b. Second solution is to leave them alone and let nature take it's course. Allow mosquitoes to go around unchecked and the ecology unchanged.
   c. Third option is to target specific species of mosquitoes that carry the malaria genes, but try to make them resistant to malaria instead of eradicating them as a whole. This seems the most reasonable common ground amongst all stakeholders.
4. Pick one solution and assess possible outcomes:
   a. Considering the information gathered, it makes more sense to adopt option 3, to attempt to instead of attempt a population collapse, which has many adverse effects and difficult to determine the scope of them, to instead attempt to eradicate the disease at the source. To make the species of mosquitoes immune or resistant to the parasite that causes malaria. This would require coordinating with many experts and legal officials from the lands this would affect the most.
   b. 3P Filter:
      i. I would be comfortable if my decision was made public, yes.
      ii. I would be very happy to present this to my parents.
      iii. I would sleep very happily at night knowing I was not only trying to save the human lives, but also that of our ecology.

    c. We can consider other solutions, but each would either make me neglect the humans currently facing disease and death, or face an entire ecosystem I can barely grasp the scope of the impacts we could accidentally cause.
5. Implement, decline, or inform (choose one):
    a. Here I would implement the best solution, which is attempting to help eradicate the diseases from the source. The parasites on the mosquitos, not the mosquitoes themselves.
6. Reflect:
    a. I think things turned out for the best. All stakeholders had a positive outcome.
    b. Follow up actions include:
        i. Ensuring the solution remains resolved and the parasite does not come back via another species, or comes back resistant to the alterations we made originally.
        ii. Also, we need to ensure that this does not cause the mosquito population to bloom if the parasite allows them to live longer, or produce more offspring. This could also alter the ecological impact of the solution provided.
    c. A couple performance metrics should be used:
        i. Number of malaria or mosquito related deaths per year should decrease.
        ii. Population size of the mosquitoes should not change by more than normal fluctuations.
    d. Key lesson learned is we should always attempt to make all life better, not just our own. We never know how big of an impact our changes can make upon the world, and usually until it is too late to reverse the effects.

Conclusion:

This ethical study provides a delicate balance of ensuring that as much life is saved as possible, including both humans and animals alike. Trying to find a true solution here is impossible due to the number of unknowns that we have, such as not being able to scope out the consequences of our actions properly. But we know that inaction will continue to see the number if diseases and deaths rise (CDC, 2026). So in my opinion the only wrong answer is inaction. We should do something for our people and do it quickly. We can learn from our mistakes as we go along, but taking action is the best course. My opinion is that a mixture of all solutions provided in the video is best, controlling a group of mosquitoes

genetics to eventually be resistant to the parasites causing the diseases, but also having a way of reversing the effects in case the mosquito populations get out of control and either collapse on their own or balloon to unmaintainable proportions.

References:

Fighting the world's deadliest animal | global health | CDC. Centers for Disease Control an Prevention. (2026, February 20). https://www.cdc.gov/global-health/impact/fighting-the-worlds-deadliest-animal.html

Bhattacharya, S., Pal, S., & Acharyya, D. (2016). *Are mosquitoes "a necessary evil"? International Journal of Fauna and Biological Studies, 3*(1, Part B), 6–11. https://www.faunajournal.com/archives/2016/vol3issue1/PartB/3-1-6.1.pdf

Appendix:

I found most surprising my desire to save people while also protecting the directly and indirectly affected wildlife. It really makes me think about how we solve these moral case studies really puts to show how much we value human life over other possible impacts we could have. I guarantee that a company such as Orkin would have a completely different opinion on how to resolve the malaria problem a large number of people are facing. I found it difficult to think from other perspectives and I tended to keep my own moral compass with me. In retrospect I think this guides (or skews) my moral decision making in ways that I am happy about, but I have no way of knowing are correct.